Finite Difference Method

Prof. Dr. Ademir João Camargo

The finite difference method is a mathematical procedure for obtaining numerical solutions to differential equations. The basic idea is to use the Taylor series to obtain formulas for the derivatives of a function. Consider the Taylor expansion of the function $f(x)$ about the point x_0:

$$f(x) = \sum_{k=0}^{n} \frac{1}{k!} \frac{d^k f(x_0)}{dx^k} (x - x_0)^k + R_n(x)$$

$$= f(x_0) + \frac{1}{1!} f'(x_0)(x - x_0) + \frac{1}{2!} f''(x_0)(x - x_0)^2 \quad (1)$$

$$+ \frac{1}{3!} f^{(3)}(x_0)(x - x_0)^3 + \cdots + \frac{1}{n!} f^{(n)}(x_0)(x - x_0)^n + R_n(x)$$

where $R_n(x)$ represents the error incurred by truncating the series after the nth-order term, and it can be estimated using the Lagrange form of the remainder:

$$R_n(x) = \frac{f^{(n+1)}(\xi)(x - x_0)^{n+1}}{(n+1)!}. \quad (2)$$

ξ is a number lying in the interval (x, x_0). Setting $x - x_0 = h$, we have $x = x_0 + h$. Substituting these values into (1), we obtain

$$f(x_0 + h) = f(x_0) + \frac{1}{1!} f'(x_0)h + \frac{1}{2!} f''(x_0)h^2 + \frac{1}{3!} f'''(x_0)h^3 + O(h^4) \quad (3)$$

Replacing h with $-h$, we can write (3) as

$$f(x_0 - h) = f(x_0) - \frac{1}{1!} f'(x_0)h + \frac{1}{2!} f''(x_0)h^2 - \frac{1}{3!} f'''(x_0)h^3 + O(h^4) \quad (4)$$

where h is the increment applied to the variable x. Using Equations (3) and (4), we can compute $f'(x)$ in three different ways, each as a difference quotient plus an error term obtained by neglecting the remaining terms of the Taylor expansion. These three methods are known as *forward, backward, and central finite differences*. The forward finite difference formula is obtained from Equation (3) by solving for $f'(x_0)$ and neglecting the terms of order $O(h^2)$ and higher, that is,

$$f'(x_0) = \frac{f(x_0 + h) - f(x_0)}{h} + O(h), \tag{5}$$

where $O(h)$ is the (first-order) truncation error incurred by neglecting the remaining terms of the univariate Taylor expansion. Differentiating (5) once more and using (5) again in the resulting expression, we obtain a formula for $f''(x_0)$, that is,

$$\begin{aligned} f''(x_0) = \frac{f'(x_0 + h) - f'(x_0)}{h} &= \frac{\frac{f(x_0 + h + h) - f(x_0 + h)}{h} - \frac{f(x_0 + h) - f(x_0)}{h}}{h} \\ &= \frac{f(x_0 + 2h) - 2f(x_0 + h) + f(x_0)}{h^2}. \end{aligned} \tag{6}$$

Repeating this procedure recursively, we arrive at the general formula for the nth derivative using the forward finite difference method:

$$f^{(n)}(x_0) = \frac{\sum_{k=0}^{n} (-1)^k \binom{n}{k} f(x_0 + (n - k)h)}{h^n}. \tag{7}$$

The binomial coefficient $\binom{n}{k}$ is given by

$$\binom{n}{k} = \frac{n!}{k!\,(n - k)!}. \tag{8}$$

The *backward finite difference* is obtained from Equation (4) by a procedure like the one used for the *forward finite difference*, that is,

$$f'(x_0) = \frac{f(x_0) - f(x_0 - h)}{h} + O(h). \tag{9}$$

Differentiating (9), we obtain

$$f''(x_0) = \frac{f'(x_0) - f'(x_0 - h)}{h} \tag{10}$$

Using (9) in (10), we obtain a formula for the second derivative using the backward finite difference method, that is,

$$\begin{aligned} f''(x_0) = \frac{f'(x_0) - f'(x_0 - h)}{h} &= \frac{\frac{f(x_0) - f(x_0 - h)}{h} - \frac{f(x_0 - h) - f(x_0 - h - h)}{h}}{h} \\ &= \frac{f(x_0) - 2f(x_0 - h) + f(x_0 - 2h)}{h^2}. \end{aligned} \tag{11}$$

By applying (9) recursively, we arrive at the general formula for the nth derivative of $f(x_0)$ using the backward finite difference method:

$$f^{(n)}(x_0) = \frac{\sum_{k=0}^{n} (-1)^k \binom{n}{k} f(x_0 - kh)}{h^n}. \tag{12}$$

To derive the central finite difference method from Equations (3) and (4), it is common in the literature to replace $h \to h/2$, that is,

$$f\left(x_0+\frac{h}{2}\right)=f(x_0)+\frac{1}{1!}f'(x_0)\left(\frac{h}{2}\right)+\frac{1}{2!}f''(x_0)\left(\frac{h}{2}\right)^2+\frac{1}{3!}f^{(3)}(x_0)\left(\frac{h}{2}\right)^3+O\left(\left(\frac{h}{2}\right)^4\right) \tag{13}$$

$$f\left(x_0-\frac{h}{2}\right)=f(x_0)-\frac{1}{1!}f'(x_0)\left(\frac{h}{2}\right)+\frac{1}{2!}f''(x_0)\left(\frac{h}{2}\right)^2-\frac{1}{3!}f'''(x_0)\left(\frac{h}{2}\right)^3+O(h^4) \tag{14}$$

Subtracting (14) from (13), we obtain the formula for $f'(x_0)$ using the *central finite difference* method:

$$f'(x_0)=\frac{f(x_0+h/2)-f(x_0-h/2)}{h}+O(h^2). \tag{15}$$

Note that in formula (15) the error is of second order; therefore, among the three first-derivative formulas, it has the smallest error.

To obtain $f''(x_0)$ using the central finite difference method, we follow a procedure analogous to those described above, that is,

$$\begin{aligned} f''(x_0)&=\frac{f'\left(x_0+\frac{h}{2}\right)-f'\left(x_0-\frac{h}{2}\right)}{h}\\ &=\frac{\frac{f\left(x_0+\frac{h}{2}+\frac{h}{2}\right)-f(x_0)}{h}-\frac{f(x_0)-f\left(x_0-\frac{h}{2}-\frac{h}{2}\right)}{h}}{h}\\ &=\frac{f(x_0+h)-2f(x_0)+f(x_0-h)}{h^2}. \end{aligned} \tag{16}$$

By successively applying this recursive procedure, we can generalize (16) to derivatives of order n using the central finite difference method:

$$f^{(n)}(x_0)=\frac{\sum_{k=0}^{n}(-1)^k\binom{n}{k}f\left(x_0+\left(\frac{n}{2}-k\right)h\right)}{h^n}. \tag{17}$$

Exercises

Find the first five terms of the Taylor series of the function $f(x)=\sin(x)$ about the point $x_0=\frac{\pi}{2}$.

Compute the first derivative of the function $f(x)=cosx$ at the point $x_0=3.0$ using the forward, backward, and central finite difference methods. Compare the results with the exact value.